Analyzing the Impact of ASR Transcription Errors on Downstream Question Answering for Egyptian Arabic Speech

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***Abstract*—This paper presents an end-to-end Speech-to-Text Question Answering (STT-QA) system for Egyptian Arabic speech using the NileTTS dataset. The system converts speech into text using Whisper-based automatic speech recognition (ASR), generates extractive question-answer pairs from reference transcripts using a local Qwen large language model, and evaluates downstream QA performance on both reference transcripts and ASR-generated transcripts. Three ASR settings were studied: a base Whisper model, an externally fine-tuned Whisper-small model by MAdel, and our own Whisper-small model fine-tuned on the NileTTS training split. The best ASR result was obtained by our fine-tuned Whisper-small model, reaching 0.1168 WER and 0.0389 CER on the evaluation split. For QA, a generative AraT5 baseline was compared with an extractive AraELECTRA model. AraT5 achieved 0.1547 F1 on reference contexts and 0.1469 F1 on ASR contexts, while AraELECTRA improved performance to 0.5558 reference F1 and 0.4858 ASR F1. The results show that ASR errors reduce downstream QA performance, but model architecture alignment with span-based QA data has a larger impact.**

***Index Terms*—Egyptian Arabic, automatic speech recognition, Whisper, Whisper-small, Qwen, AraT5, AraELECTRA, NileTTS, question answering, word error rate, exact match, F1-score.**

I. Introduction

Spoken question answering is an important step toward voice-based intelligent systems, especially for dialectal Arabic where users naturally speak in informal forms rather than write standardized Modern Standard Arabic. Egyptian Arabic presents additional challenges because speech recognition must handle dialectal pronunciation, casual wording, and domain-specific vocabulary before the question answering model can produce reliable answers.

This project implements an end-to-end Speech-to-Text Question Answering pipeline using the NileTTS dataset. The system first transcribes Egyptian Arabic speech using ASR models, then uses the resulting transcript as context for a QA model. The central research question is how ASR transcription errors affect downstream QA performance when the same questions and gold answers are evaluated using clean reference transcripts versus noisy ASR-generated transcripts.

The project contains three main stages. First, multiple Whisper-based ASR settings are evaluated on the NileTTS evaluation split. Second, Qwen is used to generate extractive QA pairs from reference transcripts because the original NileTTS dataset does not provide QA labels. Third, two QA architectures are compared: AraT5 as a generative baseline and AraELECTRA as an extractive QA model.

II. Related Work

Whisper is a multilingual encoder-decoder Transformer model for speech recognition and speech translation. Its architecture converts audio into log-Mel spectrogram features, encodes the acoustic sequence using Transformer layers, and decodes the transcript token by token. This makes it suitable for multilingual ASR tasks, including Arabic speech recognition.

AraT5 is an Arabic text-to-text Transformer model that can formulate NLP tasks as conditional generation. In this project, AraT5 is used as a generative QA baseline where the model receives a question and context and generates an answer string. However, the generated QA dataset used in this project is extractive, meaning that each gold answer is expected to appear as a span inside the transcript.

Extractive QA models are better aligned with this setup because they predict the start and end position of the answer span in the context. Therefore, the final improvement uses AraELECTRA, an Arabic extractive QA model based on a SQuAD-style question answering setup.

III. Dataset and Preprocessing

The NileTTS dataset contains Egyptian Arabic audio recordings with reference transcripts and metadata. The evaluation set used in this project contains 950 audio examples. The domains include sales, general, and medical conversations.

Arabic text normalization was applied before ASR metric calculation. The normalization process removed diacritics, removed tatweel, unified Alef variants, normalized Ya and Ta Marbuta variants, removed punctuation, and collapsed repeated whitespace. This reduced the effect of superficial Arabic spelling variation when computing WER and CER.

Since NileTTS does not contain ready-made QA labels, Qwen was used to generate QA pairs from the reference transcripts. The generated pairs were filtered to keep only examples where the answer appears inside the reference transcript. This created an extractive QA dataset suitable for both generative and extractive QA experiments.

TABLE I
QA DATASET SUMMARY

Split	QA Pairs	Unique Audios
Train	12,877	5,962
Validation	1,398	665
Test	1,554	738

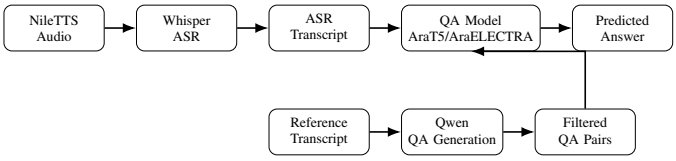


Fig. 1. End-to-end STT-QA pipeline.

IV. PROPOSED SOLUTION

A. Overall Pipeline

The complete system follows a speech-to-text-to-QA structure. Audio is first passed to the ASR model to generate a transcript. QA pairs are generated from the reference transcripts using Qwen. Then QA models are trained on the generated QA data and evaluated using both reference and ASR transcripts.

B. ASR Component

Three ASR settings were considered. The first was a base Whisper model used without task-specific fine-tuning. This served as the baseline ASR model. The second was an externally fine-tuned Whisper-small model by MAdel, used to transcribe the evaluation data and compare against the base model. The third was our own Whisper-small model fine-tuned on the NileTTS training split and evaluated on the evaluation split. This final model achieved the best ASR performance and was used to study downstream QA behavior with ASR-generated transcripts.

C. QA Pair Generation

Qwen, a large language model, was used to generate Arabic question-answer pairs from reference transcripts. The prompt required that each answer be a short span copied directly from the transcript. The generated QA data was then filtered using answer-span validation. Any row where the answer could not be found inside the reference transcript was removed. This ensured that the dataset was appropriate for extractive QA.

D. QA Models

Two QA models were compared. The first was AraT5, a generative text-to-text model. It received the question and context and generated the answer as text. The second was AraELECTRA, an extractive QA model. AraELECTRA predicts the start and end positions of the answer span in the context, making it better aligned with the generated QA labels.

V. EXPERIMENTAL SETUP

For ASR evaluation, WER and CER were calculated after Arabic normalization. WER measures word-level transcription error, while CER measures character-level transcription error.

For QA evaluation, the same test questions and gold answers were evaluated under two conditions. In the first condition, the context was the clean reference transcript. In the second condition, the context was the ASR-generated transcript. This setup isolates the effect of ASR noise because the question and gold answer remain fixed while only the context changes.

AraT5 was fine-tuned for three epochs using Qwen-generated QA pairs. It used a maximum source length of 512 tokens and maximum target length of 96 tokens. AraELECTRA was fine-tuned as an extractive QA model using maximum sequence length 384 and document stride 128. The model was trained using start and end answer positions derived from the validated answer spans.

VI. RESULTS

A. ASR Results

Table II compares the three ASR settings. The base Whisper model produced the highest WER and CER. The MAdel fine-tuned Whisper-small model improved over the base model, showing the benefit of Egyptian-Arabic-specific fine-tuning. Our own Whisper-small model fine-tuned on the NileTTS training split achieved the best performance, with 0.1168 WER and 0.0389 CER.

TABLE II
ASR RESULTS ON NILETTS EVALUATION DATA

ASR Model	WER	CER
Whisper Base	0.5363	0.1726
MAdel Fine-tuned Whisper-small	0.3721	0.1573
Our Fine-tuned Whisper-small on NileTTS	0.1168	0.0389

B. QA Model Results

Table III compares AraT5 and AraELECTRA on reference transcripts and ASR transcripts. AraT5 achieved low performance because it is a generative model and often generated answers that were not exact spans from the context. AraELECTRA achieved much stronger results because it directly predicts an answer span inside the context.

TABLE III
QA MODEL RESULTS ON REFERENCE AND ASR CONTEXTS

Model	Ref EM	Ref F1	ASR EM	ASR F1	F1 Drop
AraT5	0.0051	0.1547	0.0026	0.1469	0.0078
AraELECTRA	0.2876	0.5558	0.1622	0.4858	0.0700

C. AraELECTRA Domain Results

Table IV shows the domain-level F1 results for AraELECTRA. The model performs strongly across all domains on reference transcripts, but F1 drops when ASR transcripts are used. This confirms that transcription errors affect downstream QA performance.

TABLE IV
ARAELECTRA DOMAIN-LEVEL QA F1 RESULTS

Domain	Ref F1	ASR F1	Drop	Ans. in ASR
General	0.540	0.487	0.054	0.541
Medical	0.550	0.502	0.048	0.585
Sales	0.566	0.481	0.085	0.472

VII. DISCUSSION AND ERROR ANALYSIS

The ASR results show that fine-tuning has a large effect on transcription quality. The base Whisper model produced high word error rate, while the externally fine-tuned MAdel Whisper-small model reduced the error. Our own Whisper-small model fine-tuned on the NileTTS training split achieved the lowest WER and CER, confirming that task-specific fine-tuning on the target dataset improves ASR quality.

The QA results show that ASR errors reduce downstream QA performance. AraELECTRA F1 decreased from 0.5558 using reference transcripts to 0.4858 using ASR transcripts. This drop occurs because some gold answers no longer appear exactly in the ASR transcript after recognition errors, substitutions, or deletions.

The comparison between AraT5 and AraELECTRA also shows that model architecture is critical. AraT5 is generative and may produce semantically related text, but it is not constrained to copy the exact answer span. AraELECTRA is extractive and predicts answer positions inside the context, so it better matches the Qwen-generated extractive QA dataset.

VIII. CONCLUSION

This project implemented an end-to-end Speech-to-Text Question Answering system for Egyptian Arabic using NileTTS. The ASR stage compared Whisper Base, MAdel fine-tuned Whisper-small, and our own Whisper-small model fine-tuned on NileTTS. Our fine-tuned Whisper-small achieved the best transcription performance with 0.1168 WER and 0.0389 CER. The QA stage used Qwen-generated extractive QA pairs and compared AraT5 with AraELECTRA. AraELECTRA significantly outperformed AraT5, achieving 0.5558 F1 on reference transcripts and 0.4858 F1 on ASR transcripts. The results show that ASR errors affect downstream QA, but using an extractive QA model aligned with span-based supervision is the most important improvement.

Future work should include human review of generated QA pairs, improved dialect-specific QA fine-tuning, and semantic answer evaluation beyond exact span matching.

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